

ICAART 2026 - Paper #90

Paper Title: *Robustness of Bayesian Neural Networks Design Choices Against Weight and Bias Perturbation Induced by Single Event Upsets (SEUs)*

Reviewer #1

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 4
Originality (Newness of the ideas expressed): 3
Technical Quality (Theoretical soundness/methodology): 1
Significance (Is the problem worth the given attention?): 3
Presentation (Structure/Length/English): 3
Overall Rating (Weighted value of above items): 2

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes
Needs more experimental results? Yes
Needs comparative evaluation? Yes
Improve critical discussion ? (validation): Yes
Figures are adequate ? (in number and quality): Yes
Conclusions/Future Work are convincing? No
References are up-to-date and appropriate? Yes
Paper formatting needs adjustment? Yes
Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

This study investigates how well Bayesian Neural Networks (BNNs) withstand Single Event Upsets (SEUs). While robustness to input perturbations has been extensively examined in both deterministic and Bayesian CNNs, far less attention has been given to disturbances affecting weights and biases, especially in Bayesian CNN architectures. To bridge this gap, the authors evaluate BNN behavior in harsh conditions by simulating SEU injections and analyzing which design choices lead to greater resilience. Single Event Effects (SEEs) occur when a single ionizing particle strikes and disrupts sensitive regions within an electronic device. A specific type of SEE is the Single Event Upset (SEU). In line with prior research, this study models SEUs by injecting bit flips into the network's weights and biases while it performs inference on new data. Both deterministic and Bayesian CNNs were tested under SEUs applied to various layers, parameters, bit positions, and neuron locations. The findings indicate that deterministic CNNs achieved higher initial accuracy, but BNNs demonstrated superior robustness. Overall, BNNs produced more consistent confidence estimates, making them promising candidates for safety-critical domains.

The authors propose a new measure (Aggregate Robustness Index, ARIn) to evaluate model robustness, based on two measures from previous work (Absolute Accuracy Difference and Softmax Difference). This new measure

is essentially a weighted sum of the two previous ones. Why is this new measure necessary, and why is it calculated in this way?. This is not addressed in the paper. The authors simply state the following: "To make the model comparison easier, we devise a new aggregate robustness measure called the Aggregate Robustness Index". The motivation for this new measure must be clearly justified.

My main concern is that the experimental evaluation presented in this paper is far too limited, as it relies on only a single dataset (Ships in Satellite Imagery) to support its conclusions. To draw meaningful and robust insights, the study should incorporate multiple datasets; otherwise, the findings remain dataset-specific and cannot be considered conclusive. In addition, the models are trained only once using an 80/20 train-test split. A single training run is insufficient to reliably assess model performance. A more rigorous approach would be to employ k-fold cross-validation, using for instance $k = 10$, to compute performance metrics averaged over multiple training iterations. This would provide a more stable and representative estimate of each model's accuracy. This point is particularly important because it is well established that model performance can vary substantially depending on the initial training conditions, random seed, or stochastic optimization path. As a result, conclusions drawn from a single training session may be misleading, especially when comparing performance differences with and without perturbations. A repeated, systematic evaluation is essential to ensure that the reported differences are genuine and not artifacts of a single experimental run.

On the other hand, the results are not entirely conclusive because, for example, Table 2 shows that Deterministic Neural Network (DNNs) are more robust when considering the accuracy metric (AAD). Therefore, it is questionable whether, even though the DNN is worse than the BNN for the ASD metric, it can be concluded that it is a less robust model. Although an additional metric (ARIn) is used, it is essentially a weighted average of the squares of AAD and ASD, and since the AAD metric has a smaller magnitude than the ASD, it is less relevant to ARIn. Therefore, even for the single dataset used, the conclusions are questionable.

Therefore, the work must be substantially improved in its experimental part to strengthen the reliability and generalizability of the results. The experimental study should be expanded to include several datasets beyond Ships in Satellite Imagery. Evaluating the proposed methods on diverse data types would help determine whether the observed behaviors are consistent or dataset dependent.

Minor comments:

- Page 1, column 1, last line: typo in "reducing bandwid".
- Page 4, column 1: typo in "The model will is trained on the dataset ...".
- Page 4, equation (5) it is out of bounds of the page. This must be corrected.
- Figures 3 and 4 are not cited in the text. All the figures and tables must be cited in the text.
- Page 5, column 2, last paragraph: In the case of the Bayesian CNN, there appears to be a contradiction in the number of SEU injections, since at the beginning of the paragraph it is mentioned that 15,456 SEUs are injected, but in the following sentence it is indicated as 13,104. Please, clarify these data.
- Table 1 is not cited in the text, making it difficult to analyze. Does it show the mean results for each of the four models, considering all variations of activation functions, etc., in each model?. If so, it would also be interesting to show the standard deviation to illustrate the variability.
- Table 3, 4 and 5 are out of bounds.

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Reviewer #2

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 3

Originality (Newness of the ideas expressed): 1

Technical Quality (Theoretical soundness/methodology): 1

Significance (Is the problem worth the given attention?): 2

Presentation (Structure/Length/English): 1

Overall Rating (Weighted value of above items): 2

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Too long

Needs more experimental results? Yes

Needs comparative evaluation? Yes

Improve critical discussion ? (validation): Yes

Figures are adequate ? (in number and quality): No

Conclusions/Future Work are convincing? No

References are up-to-date and appropriate? No

Paper formatting needs adjustment? Yes

Improve English? Yes

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

The paper describes using a Bayesian CNN to select a task – the novelty of the proposal is not described and what more is needed – a proper justification for such a solution in 2025 – the background also does not describe the current state of the art and problem formalization. The experimental section also does not show detailed ablation studies, hyperparameter analysis or comparison with state of art.

In general, the paper describes a solution for practical application.